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Q2	
Q3	
Q4	
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University of Canterbury

Mid-Year Exam 2025

Prescription Number(s):	**ENEL445**
Paper Title:	**Applied Engineering Optimisation**

Time allowed: **2 hours**

Number of pages: **20**

Calculators **ARE** permitted.

ALL Questions **ARE COMPULSORY**.
(There are 4 questions in total.
The maximum total score is 80 marks.)

Family name (in block letters): ....

Given name: ....

Student number: ....

INSTRUCTIONS

This is an closed-book test (apart from the A4 cheat sheet). Please write your answers in the space provided. There is extra space at the end of the test should you need it. Show your working as you complete the test, so that you can more easily receive partial credit. Make a note indicating when your work is continued on an extra page.

Q1. Gradient-free Optimisation Part I

(a) [2 marks]

Name, or define, **one** optimisation problem for which gradient-free methods are suited to solve, and give **one** reason why this is the case.

(b) [2 marks]

A contour plot for an objective function is shown in Figure 1 below. We want to **maximise** this function. On Figure 1, carefully sketch the **outside contraction** operation for the Nelder-Mead method. No need to continue to complete the iteration.

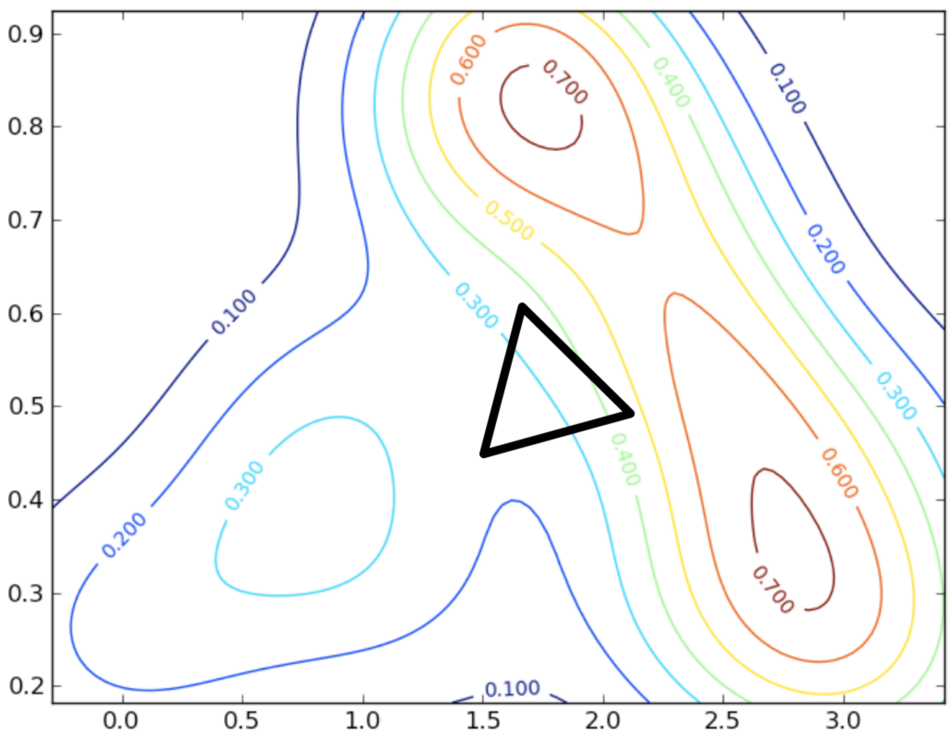

Figure 1: Contour plot for question 1(b)

(c) **[2 marks]**

Briefly explain how a generalised pattern search algorithm works.

(d) **[2 marks]**

Define Lipschitz continuity and explain how Shubert's algorithm utilises this property to find the minimum of a function. Feel free to draw pictures.

(e) **[2 marks]**

On the f-d plot below, circle the dots representing the rectangles that will be subdivided next by the DIRECT method. The value for $\epsilon|f_{\min}| = 0$. Assume all point-shifts have been done.

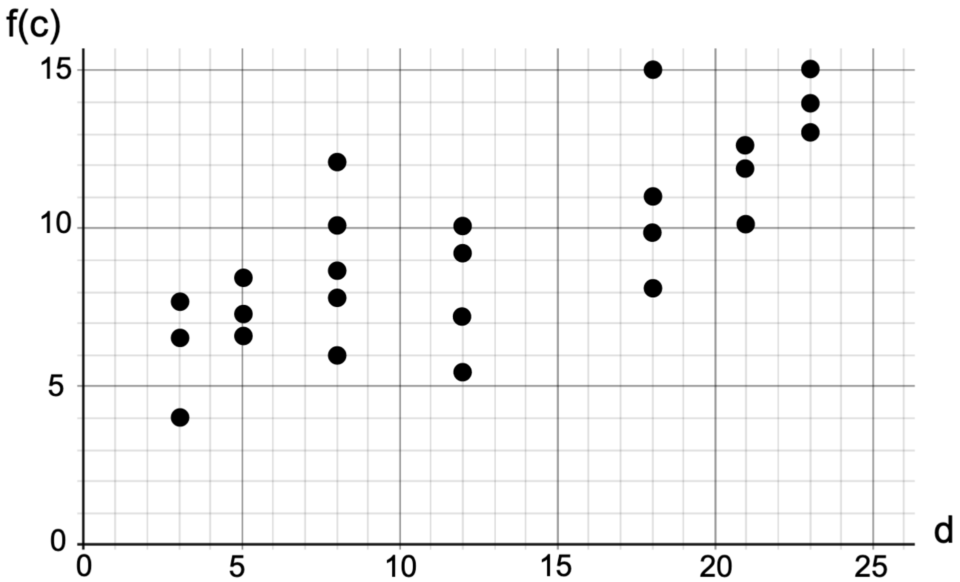

Figure 2: f-d plot

(f) **[3 marks]**

State what c, $f(c)$, and d are on the axes of the plot in Figure 2 **AND** explain the purpose of the ϵ parameter and why might setting $\epsilon = 0$ be undesirable.

(g) **[1 mark]**

What is elitism in a genetic algorithm?

(h) [3 marks]

The update rule for the particle swarm algorithm is given by

$$x_{k+1}^{(i)} = x_k^{(i)} + v_{k+1}^{(i)} \Delta t \tag{1}$$

$$v_{k+1}^{(i)} = \alpha v_k^{(i)} + \beta \frac{x_{\text{best}}^{(i)} - x_k^{(i)}}{\Delta t} + \gamma \frac{x_{\text{best}} - x_k^{(i)}}{\Delta t} \tag{2}$$

What does $x_k^{(i)}$, $v_k^{(i)}$, $x_{\text{best}}^{(i)}$, and x_{best} represent? Make sure you define the meaning of all symbols such as k and i.

(i) [3 marks]

In Figure 3 below, draw a point to indicate the position of $x_{k+1}^{(i)}$ showing any intermediate vectors that were needed to arrive at that point. Assume $\alpha = \beta = \gamma = 1$ in Equation (2).

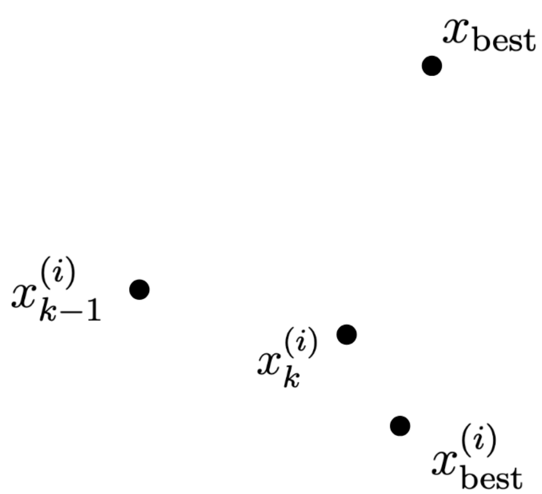

Figure 3: Current state of particles.

Q2. Gradient-free Optimisation Part II

(a) [5 marks]

Convert the following linear programming problem into standard form

$$\begin{array}{ll} \text{maximise} & -x_1 - 5x_2 + 2x_3 \\ \text{subject to} & x_2 - x_3 \geq 7 \\ & \left|x_1 + 4x_2 - x_3\right| \geq 1 \\ & x_1 \geq 0, \ \ x_2 \text{ free}, \ \ x_3 \geq 0 \end{array}$$

(b) [4 marks]

Solve the following optimal assignment problem (find the assignment with the least cost) using the Hungarian algorithm:

	Task 1	Task 2	Task 3
Worker A	145	165	190
Worker B	190	175	215
Worker C	160	180	260

(c) [2 marks]

What does it mean for a transportation problem to be balanced? **AND** what is the additional constraint that is needed for a balanced transportation problem to be feasible?

(d) [3 marks]

Briefly explain the simulated annealing algorithm, incorporating the terms **acceptance probability** and **cooling schedule**.

(e) [3 marks]

Briefly explain the difference between a depth-first and a breadth-first strategy in the branch-and-bound algorithm **AND** state an advantage of depth-first over breadth-first.

(f) [1 mark]

Explain what is meant by a Nash equilibrium.

(g) [2 marks]

You are trying to minimise a function $f(\mathbf{x})$, but each evaluation of the function takes 30 hours on the fastest computer you can afford. State an optimisation strategy that you can apply to this problem and briefly describe how this strategy works.

Q3. Unconstrained Optimisation

(a) Consider generating a generalised Van der Corput sequence for initializing an unconstrained optimization algorithm with multiple initial solutions.

(i) **[3 marks]**

Let i be a positive integer. We first need to represent i in base b as

$$i = a_0 + a_1 b + a_2 b^2 + \cdots + a_r b^r, \tag{3}$$

where $0 < a_r \leq b - 1$ and $0 \leq a_j \leq b - 1$ for $j = 0, 1, 2, ..., r - 1$. In other words, i is expressed as a $(r + 1)$-digit, $(a_r a_{r-1} \cdots a_0)_b$, in base b.

Apply (3) to find the representation of $i = 2, 3, 4, 5$ in base $b = 3$.

(ii) **[3 marks]**

The ith element in the generalized Van der Corput sequence can then be computed using the following radical inversion function:

$$\phi_i(b) = \frac{a_0}{b} + \frac{a_1}{b^2} + \cdots + \frac{a_r}{b^{r+1}}. \tag{4}$$

Use your results in the previous question and (4) to find $\phi_i(3)$ for $i = 2, 3, 4, 5$.

(b) Consider the following unconstrained optimization problem:

$$\min_{\mathbf{x}} L(\mathbf{x}) = f(\mathbf{x}_0) + \triangledown f(\mathbf{x}_0)^T(\mathbf{x} - \mathbf{x}_0) + \frac{1}{2\mu}(\mathbf{x} - \mathbf{x}_0)^T(\mathbf{x} - \mathbf{x}_0), \tag{5}$$

where $\mathbf{x}$ is the design variable vector, and $\mu > 0$ is a constant scalar. $\mathbf{x}_0$ is **known**. $f(\mathbf{x}_0)$ is the value of the function $f(\cdot)$ at $\mathbf{x}_0$ and $\triangledown f(\mathbf{x}_0)$ is the gradient of the function $f(\cdot)$ at $\mathbf{x}_0$. $f(\mathbf{x}_0)$ and $\triangledown f(\mathbf{x}_0)$ thus also have **known** values.

(i) **[4 marks]**

Compute the gradient $\frac{\partial L(\mathbf{x})}{\partial \mathbf{x}}$ and the Hessian $\frac{\partial^2 L(\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}^T}$.

(ii) **[4 marks]**

Set the gradient $\frac{\partial L(\mathbf{x})}{\partial \mathbf{x}}$ derived in the previous question to zero to find a solution to the minimization problem in (5).

Justify that the obtained solution is at least a local optimal solution.

(iii) [6 marks]

If the constant μ in (5) is negative, i.e., $\mu < 0$, can we still find a bounded solution to (5)? Explain your answer.

Here, a solution is bounded if $||\mathbf{x}||_2 < +\infty$, where $||\cdot||_2$ is the 2-norm.

Q4. Power systems optimisation & algorithms

(a) [6 marks]

Why is it computationally difficult to solve the AC optimal power flow problem for a large power system? What can be done to mitigate this (explain every way you are aware of)?

(b) [4 marks]

Explain the concept of *security-constrained* AC optimal power flow (SC-OPF). Why is SC-OPF hard to solve in practice?

(c) [5 marks]

Consider the three-bus system depicted in Fig. 4. You may assume that each meter is very close to the nearest bus in the figure.

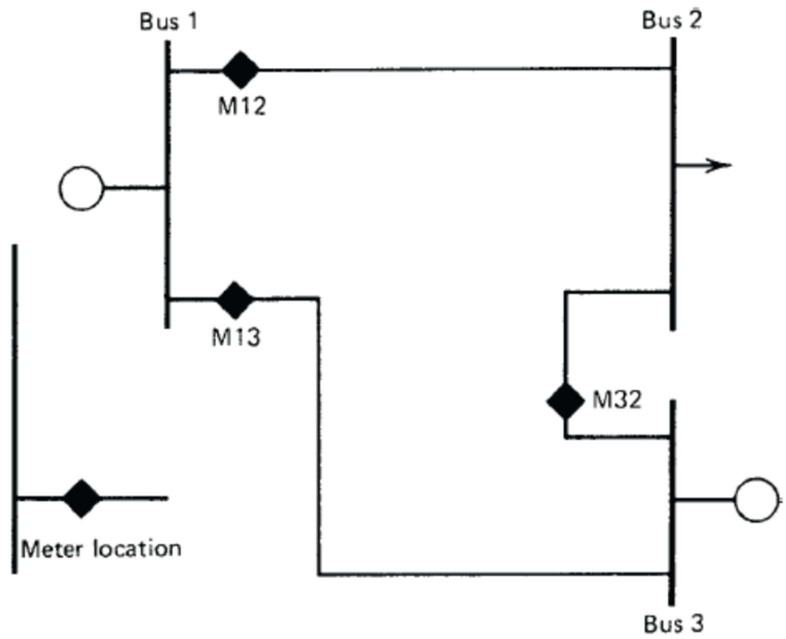

Figure 4: Power system for question 4 (c,d)

Given (potentially noisy) measurements of P_{12}, P_{13}, P_{32} (i.e. the power-flows of every line), **formulate a *linear* least squares problem to solve for the power system states (voltage angles at each bus)**. You may assume that all meters have the same standard deviation (in per unit).

The active power flow through the line from i to j may be approximated by:
$P_{ij} = B_{ij}(\theta_i - \theta_j)$.

Hint: You do NOT need to solve this problem - simply write the optimisation problem to be solved in a standard form, or write the form of the solution if you prefer. It is important to define any matrices, vectors, and variables.

(d) [5 marks]

Suppose reactive power measurements are available at each meter, (potentially noisy) voltage magnitude and (synchronized) angle measurements are available at buses 2 and 3, and B_{12}, B_{13} are known (while B_{32} is completely unknown). **Formulate a *non-linear* least squares problem to solve for the power system states and unknown parameters (voltage magnitudes and angles at each bus, and** B_{32}). You may ignore any discrepancies in meter accuracy or units - simply show the correct form of the problem.

The active power flow through the line from i to j is given by:
$P_{ij} = V_i V_j B_{ij} \sin(\theta_i - \theta_j)$.

The reactive power leaving bus i through the line from i to j is given by:
$Q_{ij} = V_i(V_i - V_j \cos(\theta_i - \theta_j))B_{ij}$.

Hint: Again, you do NOT need to solve this problem - simply write the optimisation problem to be solved in a standard form.

Extra Answer Space (Optional)

Extra Answer Space (Optional)

END OF PAPER